

<https://doi.org/10.1038/s41538-026-00908-2>

Isolation and characterization of *Lactobacillus plantarum* LP2301 from bovine milk: acid and bile tolerance, antibacterial activity, and genomic safety evaluation

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The long-term and non-standard use of antibiotics poses a significant threat to food safety and human health. Probiotics are a promising alternative to antibiotics, effectively combating bacterial infections while reducing the risks associated with antibiotic resistance. Gram staining confirmed LP2301 as a Gram-positive bacterium, and species identification as *Lactobacillus plantarum* was established by 16S rRNA gene sequencing and whole-genome sequencing. In this study, LP2301 was isolated from the milk of a healthy dairy cow, which exhibits characteristics of acid (pH = 2.5) and 0.3% bile salt tolerance and the ability to secrete effective substances that inhibit the growth of pathogenic bacteria. Through whole-genome sequencing, we identified plantaricin biosynthetic gene clusters (including *plnE* and *plnF*) and the absence of hemolytic ability in LP2301. Combined experimental and genomic evidence suggests that the antimicrobial activity is attributable to bacteriocins, particularly plantaricin-related peptides, and Proteinase K treatment experiment confirmed the presence of proteinaceous bacteriocin-like substances contributing to antimicrobial activity. However, the specific effective components require further experimental verification. These findings suggest that LP2301 has potential as a natural antimicrobial agent and a candidate for further evaluation as an antibiotic alternative, pending in vivo validation.

Antibiotic overuse and misuse pose a significant threat to global health through multiple interconnected mechanisms, with antimicrobial resistance (AMR) representing the most critical challenge, which promotes the emergence and proliferation of resistant bacterial strains via mechanisms such as enzymatic degradation of drugs (e.g., β -lactamases), target site modifications, efflux pumps, and horizontal gene transfer (HGT) of resistance determinants^{1–4}. These pathogens can transfer to humans through contaminated food products, direct contact, or environmental dissemination, amplifying the public health burden^{5,6}. Beyond driving resistance, antibiotics profoundly disrupt the gut microbiome, leading to dysbiosis characterized by reduced microbial diversity, depletion of beneficial commensals, and overgrowth of opportunistic pathogens such as *Clostridioides*

difficile and *Enterobacteriaceae*. This disruption compromises colonization resistance, increasing host susceptibility to infections and facilitating the expansion of antimicrobial resistance genes (ARGs) within the gut resistome^{4,7–11}. Without urgent global collaboration, AMR could cause 10 million annual deaths by 2050, underscoring the imperative for integrated, cross-sectoral responses to preserve antibiotic efficacy and safeguard public health^{12,13}.

The application of probiotics, particularly *Lactobacillus plantarum*, as alternatives to antibiotics represents a promising strategy to combat bacterial infections while mitigating the risks associated with antibiotic resistance^{14–16}. This approach leverages multiple mechanisms of action inherent to probiotics, offering a multifaceted alternative to traditional

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antimicrobial therapies. *L. plantarum* exhibits potent antimicrobial activity through the synthesis of various compounds, including bacteriocins (e.g., plantaricin), organic acids (lactic acid, acetic acid), hydrogen peroxide, and biosurfactants^{15,17,18}. These substances directly inhibit the growth and viability of a broad spectrum of pathogenic bacteria, such as *Salmonella* spp., *Escherichia coli* O157:H7, *Staphylococcus aureus*, *Listeria monocytogenes*, and *Pseudomonas aeruginosa*. Bacteriocins disrupt bacterial cell membranes and metabolic processes, while biosurfactants demonstrate significant antibacterial and antibiofilm properties, effectively reducing biofilm formation and viability of pathogens^{19–23}. The cell-free supernatants (CFS) of *L. plantarum* cultures retain potent antimicrobial activity, suggesting the potential for topical or therapeutic applications beyond direct probiotic administration^{21,24}.

Beyond direct antimicrobial effects, *L. plantarum* enhances host defense mechanisms. It strengthens intestinal barrier function by upregulating the expression of tight junction proteins (e.g., E-cadherin) and mucins (e.g., MUC1), thereby reducing pathogen translocation across the gut epithelium^{25,26}. Furthermore, *L. plantarum* demonstrates anti-oxidative properties, contributing to overall cellular health²⁵. *L. plantarum* competitively excludes pathogens by colonizing the gastrointestinal tract, adhering to intestinal epithelial cells, and producing an environment unfavorable for pathogenic growth through pH reduction and antimicrobial compound secretion. This competitive exclusion disrupts the colonization and proliferation of pathogens like *Salmonella*, ETEC K88, and *Eimeria* spp., reducing infection severity and diarrheal symptoms in animal models^{20,22,26,27}. Oral administration of *L. plantarum* can modulate the gut microbiota composition, influencing the abundance of indigenous *Lactobacilli* and potentially promoting a more resilient microbial community²⁸. *L. plantarum* functions as a multifaceted alternative to antibiotics through its ability to produce direct antimicrobial agents, enhance host barrier and immune functions, modulate the microbiome, and competitively exclude pathogens^{29–31}.

In this research, a strain of *L. plantarum*, LP2301, was successfully isolated and identified from the milk of healthy cows. Subsequently, its whole-genome sequencing, physiological and biochemical characteristics, growth curve, Gram staining, and bacteriostatic circle were examined. Notably, LP2301 is capable of efficiently secreting substances that can inhibit *Escherichia coli*, and it can withstand bile salts and acidic conditions. The findings of this study indicate that LP2301 exhibits promising probiotic properties and warrants further investigation as a potential antibiotic alternative and oral modulator of intestinal microbiota.

Results

Identification result of LP2301

The BLAST result of the 16S rRNA gene sequence (Fig. S1A) showed 99.02% identity with *Lactobacillus plantarum* strain TMPC 2M625 (Accession: OM757923.1) and 98.60% identity with *L. plantarum* TMPC 31B14 (Accession: OL423291.1). The de novo sequencing result of the pie chart of homologous species classification in the NR (non-redundant protein sequence) database is shown in Fig. S1B.

Morphological and growth characteristics of LP2301

To understand the growth status of LP2301, its growth characteristics were studied by MRS plate observation, Gram staining, and growth curve. The results revealed that LP2301 presented as milky white colonies on MRS plates, with large colonies and smooth edges (Fig. 1A). In Fig. 1B, Gram staining confirmed that it was a Gram-positive bacterium. Its growth curve was also measured, and the result indicated that LP2301 entered the exponential phase after 4 h, reached the stationary phase at approximately 12 h (OD₆₀₀ = 1.1476), and maintained this plateau until 36 h (OD₆₀₀ = 1.2646). After 36 h, OD₆₀₀ decreased slightly to 1.2458 at 48 h.

The biochemical characterization of LP2301

The biochemical characterization of LP2301 is shown in Table 1. The findings indicated that LP2301 did not generate oxidase, fructose, xylose, or

D-arabinose, whereas all the other tested indicators yielded positive results. Based on all characteristics (lactic acid bacteria, extensive sugar fermentation profile, specific sugar non-fermentation), the results strongly support its identification as *L. plantarum*. It is worth noting that while LP2301 demonstrated growth in MRS broth containing 0.3% bile salts. Subsequently, 16S rRNA gene sequencing was further conducted for final molecular identification to confirm its identity.

The bile salt tolerance and acid resistance of LP2301

A bile salt tolerance assay of LP2301 was conducted. The results showed that although there was a significant reduction in growth compared to normal medium, LP2301 could still survive in large numbers in a 0.3% bile salt environment (Fig. 2A, B). After being cultured in a 0.3% bile salt solution for 4 h, the survival rate was 91%, which remained at a high level and did not show a significant decrease. This is completely consistent with expectations. Importantly, the colony count returned to the initial level in 2 h. This indicates that the strain population exhibits robust resilience and recovery capabilities, enabling it to surmount transient pressure-induced damage.

To further verify the viability of LP2301, an acid tolerance experiment was conducted. As shown in Fig. 3A, LP2301 can grow normally in an environment with pH=2.5. Except for the significant difference in the first hour compared to the control, there were no significant differences in the number of colonies at 0 h, 0.5 h, and 2 h (Fig. 3B). Compared with the control group, after being treated with pH 2.5 for 1 h, the survival rate of LP2301 was 95%. This indicates that LP2301 can tolerate an acidic environment with pH = 2.5. The strain did not experience significant mortality upon exposure to strong acid, and its survival rate remained extremely high even after being exposed to gastric acid for 0.5 hours. The colony count at 2 h showed no difference from that at 0 h, indicating not just survival but potentially also the activation of repair mechanisms or even the resumption of growth after adapting to the acidic environment. This is crucial for probiotics to cope with the complex and dynamic environment within the stomach.

The antibacterial ability of LP2301

To verify the antibacterial properties of LP2301 as a probiotic, antimicrobial activity assay and antibacterial zone tests were executed which results were shown in Figs. 4 and S1. As shown in Fig. 4A, the number of *E. coli* on the LB plates with LP2301 supernatant added at 0 h, 2 h, 4 h, and 8 h was significantly lower than that on the LB plates without LP2301 supernatant added. Through statistical analysis, it can also be found that, except for the 4-hour time point, there were significant statistical differences in the number of *E. coli* between the two groups at the other three time points. Subsequently, the experimental results of the antibacterial zone also demonstrate that the supernatant of LP2301 exhibits a notable antibacterial effect (Fig. 4C). During the growth process of LP2301, it secretes certain metabolites with antibacterial activity. In Fig. S1C, the results indicated that the supernatant of LP2301 exhibited inhibitory effects against *Staphylococcus aureus*, *Salmonella typhimurium*, and *Bacillus cereus*. The reduction in activity after Proteinase K indicates the presence of proteinaceous bacteriocin-like substances (Fig. S1D).

The whole-genome characteristics of LP2301

To examine the genetic characteristics of LP2301 and its ability to secrete effective antimicrobial substances, whole-genome sequencing was performed. The whole genome size is 3,174,340 bp. The GC content of the LP2301 gene is 44.57%. A contig number is 66, and N50 is 139,470 bp. The genome coverage statistics are presented in Table S2. The number of predicted protein-coding genes is 2981. Figure 5A and 5B present the gene enrichment analysis conducted using gene ontology (GO) and Kyoto Encyclopedia of genes and genomes (KEGG). An analysis of the pathogen-host interactions of LP2301 was conducted. By using BLAST to align gene protein sequences with the PHI-base database (Pathogen Host Interactions Database), the results are shown in Table 2 (Top 20). The virulence factors obtained from sequencing LP2301 were also annotated, as shown in Table

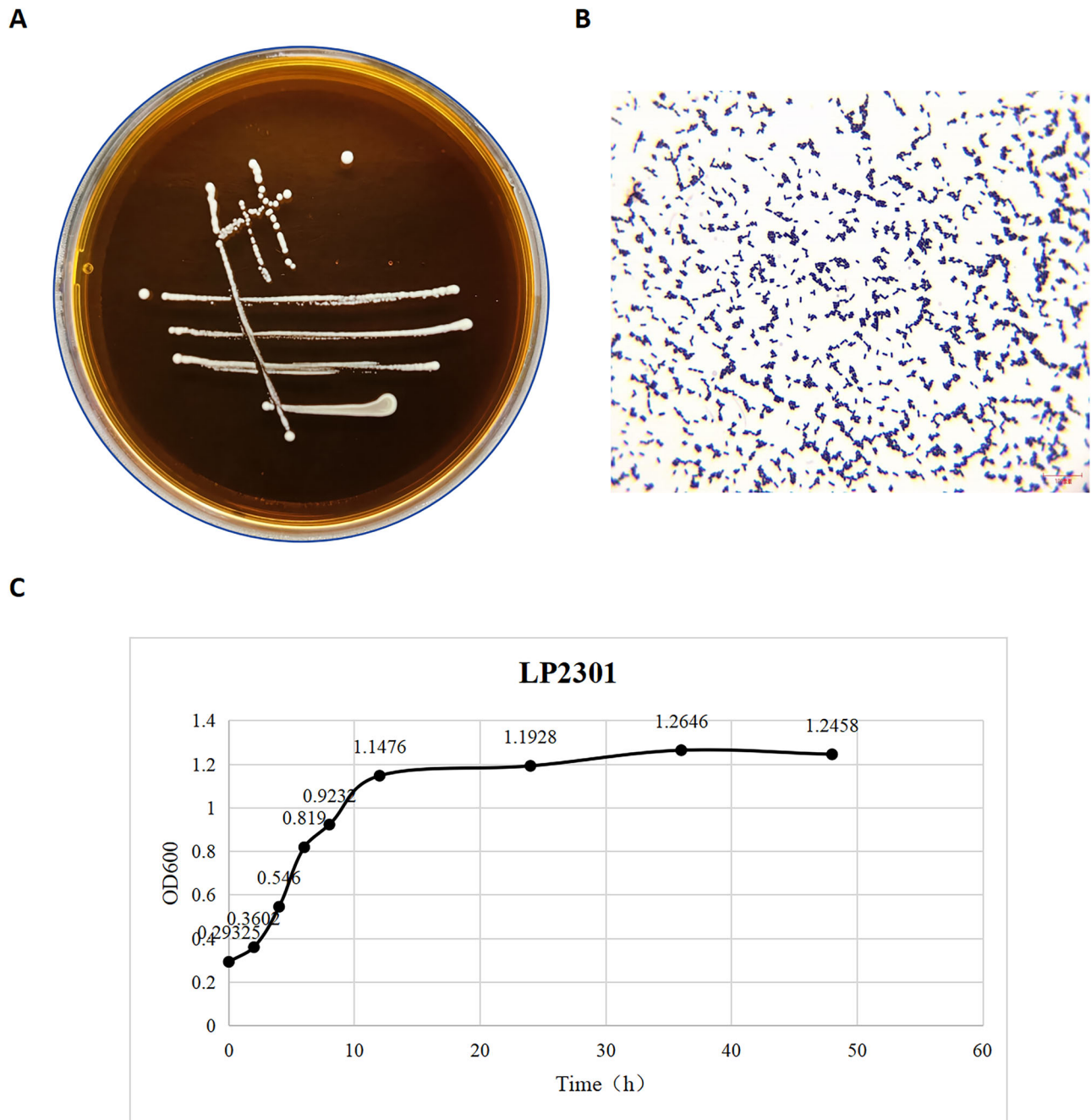


Fig. 1 | Growth characteristics of LP2301. A Growth morphology of LP2301 on MRS agar; **B** Gram staining of LP2301 (100× magnification); **C** Growth curve of LP2301 in MRS broth.

3 (Top 20). Genes with sequence similarity to those involved in pathogen-host interactions (PHI-base) and virulence factors (VFDB) were identified. These include conserved housekeeping genes (e.g., *clpC*, *clpE*, and *phoP*) that are commonly found in lactic acid bacteria and do not confer pathogenic potential. In this annotation result, no typical enterotoxin genes were found in LP2301. Meanwhile, genes such as *lap*, *clpC*, *prsA2*, *gtcA*, and *efaA*, which are related to adhesion, were detected in LP2301. This is because probiotics need to adhere to intestinal epithelial cells in order to colonize, competitively exclude pathogens, and modulate immunity. Although we have identified some genes related to hemolysis (pident 30–60%, high e-value), these homologous genes may encode completely different functions in *L. plantarum*, such as membrane transporters (ABC transporters), stress response proteins, enzymes involved in cell wall

synthesis, 3-ketoacyl-ACP reductase CylG, and biofilm formation. To confirm the absence of hemolysis in LP2301, we also tested its hemolytic properties (Fig. S1E). The results indicated that LP2301 did not form a hemolysis halo on the blood agar plate. Signal peptide-containing proteins were predicted using SignalP. LP2301 secretes proteins with signal peptides (Table 4).

Through in-depth genome mining and functional annotation analysis of transporters from second-generation sequencing, a putative plantaricin biosynthetic gene cluster was found in LP2301, comprising structural genes such as *plnE* and *plnF*, along with genes encoding immunity proteins and an ABC transporter system (Table S1). These findings support the hypothesis that the observed antimicrobial activity may be partly attributed to bacteriocin production.

Table 1 | The physiological and biochemical identification of LP2301

Experimental project	Results
Oxidase	—
Catalase	+
Fructose	—
Galactose	+
Xylose	—
D-arabinose	—
D-melibiose	+
Esculin	+
Cellobiose	+
Maltose	+
Mannitol	+
Salicin	+
Sorbitol	+
Sucrose	+
Raffinose	+
Inulin	+
Lactose	+

“+” indicate positive of secreting; “—” indicate negative of secreting.

Discussion

LP2301 was successfully isolated from bovine milk with combined acid (pH 2.5) and bile salt (0.3%) tolerance and with multi-antibacterial activity against Gram-positive and Gram-negative bacteria. Compared to *L. plantarum* LPH10, which exhibited tolerance in simulated gastric acid (pH 3.0)³², LP2301 exhibits superior acid survival at pH 2.5. Although LP2301 exhibited a generally versatile carbohydrate fermentation profile, it did not ferment fructose under the tested conditions, highlighting strain-specific metabolic diversity. The strain's acid production and sugar fermentation profile suggest potential for food fermentation applications, but direct experimental validation is required. This adaptability stems from its versatile metabolic pathways, which enable efficient carbohydrate utilization and lactic acid production, driving pH reduction and preservation while enhancing product characteristics^{33–36}.

L. plantarum exhibits remarkable acid and bile salt tolerance, which is crucial for its survival and functionality in the gastrointestinal tract. Several studies provide comprehensive insights into the mechanisms underlying this tolerance and its implications for probiotic applications. *L. plantarum* strains demonstrate significant resistance to acidic environments, as evidenced by their ability to maintain high survival rates in simulated gastric acid conditions (pH 3.0)³⁷. This acid tolerance is further enhanced when these strains are cultured in specific growth media, such as terebinth milk, where *L. plantarum* exhibited higher survival at pH 2.5 compared to standard MRS broth³⁸. The acid resistance mechanisms appear to be strain-specific and may involve complex cellular adaptations. Exposure to low pH triggers a multifaceted response in *L. plantarum*. Acid tolerance involves the induction of general stress response proteins (e.g., chaperones GroEL, DnaK) to prevent protein denaturation^{39,40}. Cells maintain intracellular pH homeostasis through proton extrusion via F₀F₁-ATP synthase and increased amino acid decarboxylation (e.g., lysine biosynthesis), consuming intracellular protons^{39,41}. Acid adaptation also remodels cell membrane composition by increasing saturated and cyclopropane fatty acids, enhancing membrane integrity and reducing permeability to protons⁴¹. The high survival rate at pH 2.5 (95% after 1 h) suggests that LP2301 has the ability of acid tolerance, which is a desirable property for probiotics intended for oral administration, although in vivo validation is required.

Regarding bile salt tolerance, comparative proteomic analyses have identified key proteins involved in bile salt response and adaptation⁴². These include glutathione reductases that protect against oxidative injury, cyclopropane-fatty-acyl-phospholipid synthase, maintaining cell envelope integrity, bile salt hydrolase (BSH), ABC transporters, and F₀F₁-ATP synthase participating in active removal of bile-related stress factors. Notably, BSH activity plays a crucial role in bile salt tolerance and has been extensively studied across various *L. plantarum* strains^{43–45}. Furthermore, several studies have demonstrated that food-associated *L. plantarum* strains possess inherent bile salt tolerance, allowing them to survive gastrointestinal transit and potentially confer health benefits^{46,47}. This characteristic is not limited to gut-associated microbes but extends to food-origin strains, indicating an evolutionarily conserved trait among *Lactobacillus* species⁴⁷. This tolerance is particularly important for their probiotic functionality, including anti-inflammatory properties, cholesterol-lowering effects, and modulation of gut microbiota^{37,48,49}.

L. plantarum supernatants exhibit antimicrobial activity against multiple clinically relevant pathogens, including Gram-negative (*E. coli* and *Salmonella*) and Gram-positive (*S. aureus*) bacteria, mediated by a complex mixture of bioactive compounds whose efficacy is influenced by cultivation conditions and strain specificity. The antimicrobial mechanism of substances secreted in the supernatant of *L. plantarum* involves a complex interplay of multiple compounds, primarily bacteriocins, organic acids (lactic, acetic, propionic, phenyllactic acid)^{18,21,50}, antifungal compounds (3-phenyllactic acid, benzeneacetic acid derivatives, hydroxy fatty acids)^{18,51,52}, and bacteriocins or antimicrobial peptides^{17,53}, which collectively disrupt pathogen viability through membrane damage, interference with cellular processes, and inhibition of biofilm formation. In this study, we speculate that the antimicrobial substance secreted by LP2301 is a bacteriocin.

Bacteriocins, such as plantaricin variants, are ribosomally synthesized peptides that exhibit activity against various Gram-positive and Gram-negative pathogens. These peptides target bacterial membranes, increasing permeability and causing leakage of intracellular components like ATP and K⁺, ultimately leading to cell death. For instance, plantaricin produced by *L. plantarum* strains destabilizes the membrane integrity of pathogens like *Salmonella* and *E. coli*, as evidenced by electron microscopy showing morphological damage^{19,54}. Organic acids, including lactic, acetic, and phenyllactic acids, contribute significantly to antimicrobial activity by lowering environmental pH and uncoupling membrane potential. The proton gradient dissipation disrupts proton motive force (PMF), reducing ATP synthesis and compromising cellular energy metabolism. For example, L-PLA from *L. plantarum* ZJ316 rapidly depletes ATP in *Salmonella* cells and increases extracellular conductivity, indicating membrane disruption^{50,55}. Collectively, these mechanisms underscore the potential of *L. plantarum* supernatant as a natural antimicrobial agent in food preservation and biomedical applications, leveraging its multifaceted approach to pathogen inhibition^{21,56}.

Although LP2301 carries genes annotated in the VFDB and PHI-base databases, these should not be interpreted as evidence of pathogenicity. Many of these genes encode conserved housekeeping functions, such as ATP-dependent proteases, ABC transporters, and two-component regulatory systems, which are common in lactic acid bacteria and may contribute to stress tolerance, niche adaptation, or interference with pathogen virulence. *L. plantarum* naturally carries multiple intrinsic resistance genes (such as *macB*, *lmrD*, and *msbA*), which are often located on the chromosome, are not transferable, and do not compromise the safety of probiotics. Many resistance genes encode efflux pumps, which may simultaneously participate in stress tolerance, bile salt resistance, intestinal colonization, and other probiotic functions. LP2301 has the core physical characteristics of excellent probiotics, which include excellent gastrointestinal viability. It can not only resist the strong killing effect of gastric acid, but also grow vigorously under bile pressure. LP2301 can secrete stable and bioactive

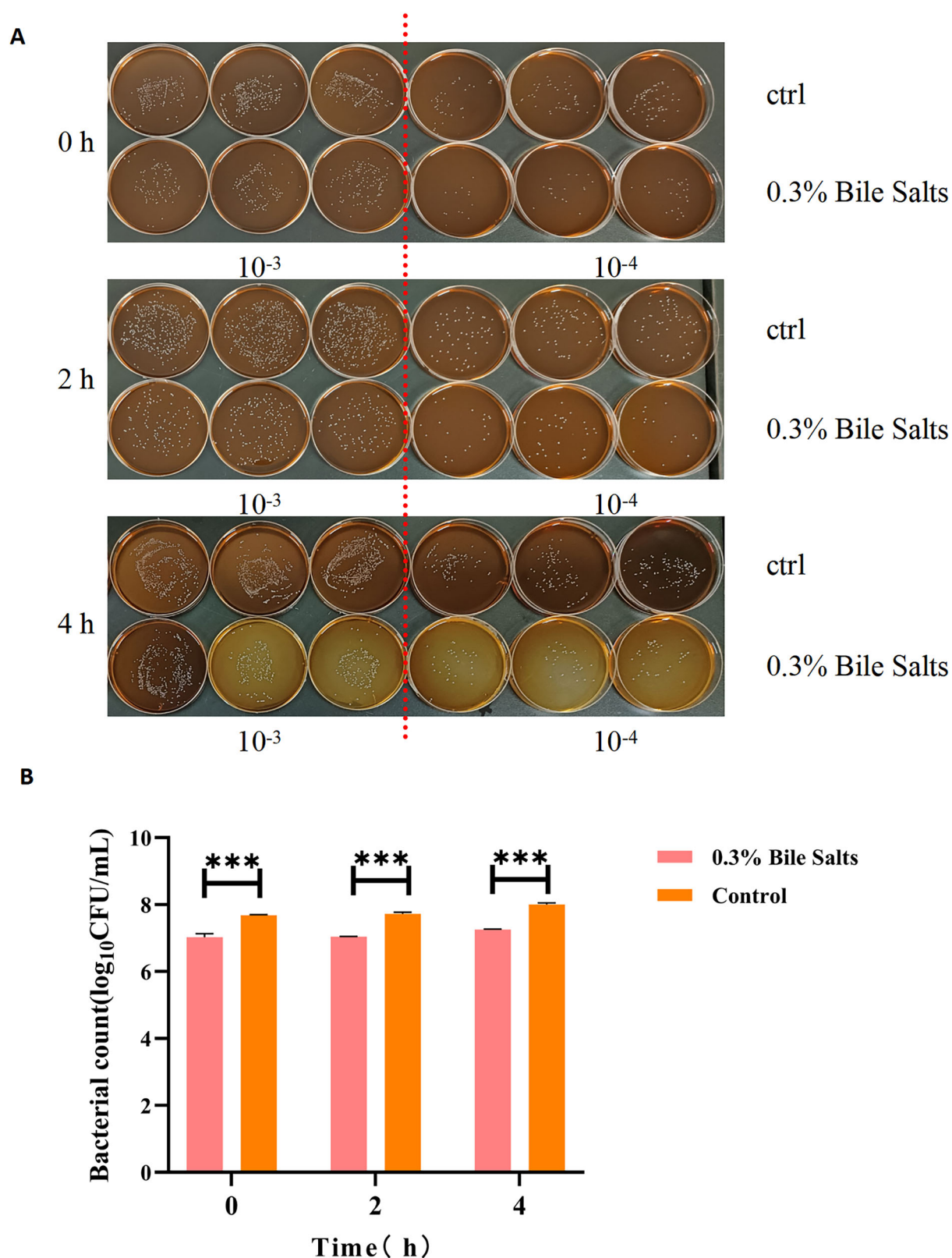


Fig. 2 | The resistance of LP2301 to bile salt. A Tolerance of LP2301 to 0.3% bile salt; **B** Statistical results of LP2301 bile salt tolerance at different time points.

metabolites and effectively inhibit the growth of various pathogenic bacteria in vitro. Limitations of this study include the lack of in vivo efficacy and safety data. Future work should include animal models to evaluate protective effects against enteric infections, comprehensive safety toxicology, and purification of the active antimicrobial peptides.

Methods

Isolation and culture of *L. plantarum*

L. plantarum 2301 (LP2301) was aseptically isolated from a single healthy lactating dairy cow's milk at a dairy cow farm (Daqing, China). In brief, healthy lactating cows were restrained, and their udder and

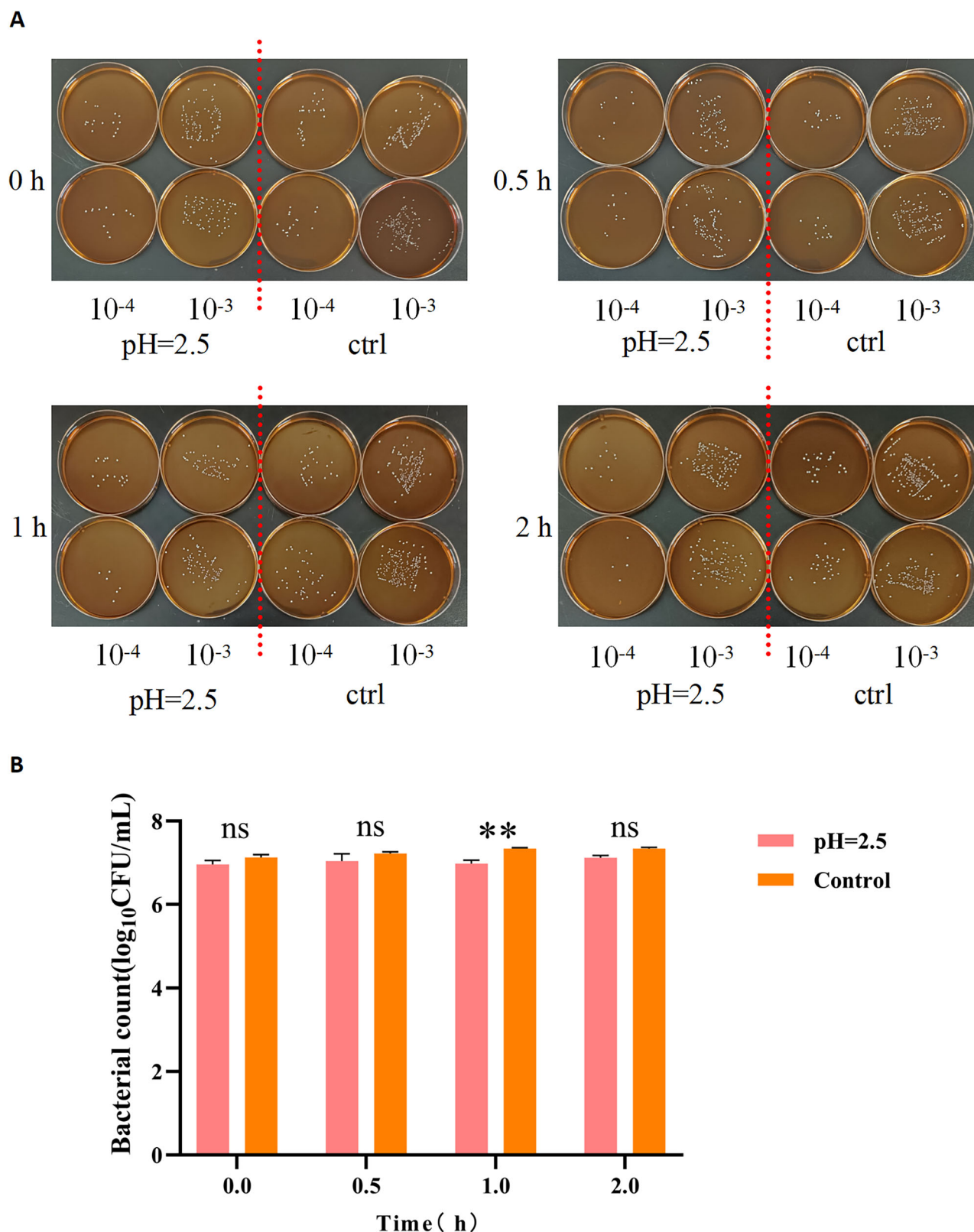


Fig. 3 | The acid resistance of LP2301. A Tolerance of LP2301 to an environment with pH 2.5; **B** Statistical results of LP2301 acid resistance at different time points.

teats were disinfected. The first three milk samples were discarded. The milk was then squeezed into sterile 50 mL tubes, sealed, and transported on ice to the laboratory within 3 hours for further processing. The milk was shaken well, and 100 μ L of milk was added to 900 μ L of sterile physiological saline, followed by thorough mixing. The original

milk solution and the diluted solution were evenly spread onto MRS agar plates, which were then inverted and incubated in a 37 $^{\circ}$ C incubator for 12–24 h. Subsequently, single colonies were incubated in a 37 $^{\circ}$ C incubator for 12–24 h for purification. The purified single colonies were then cultured in MRS broth for 12–24 h until the broth

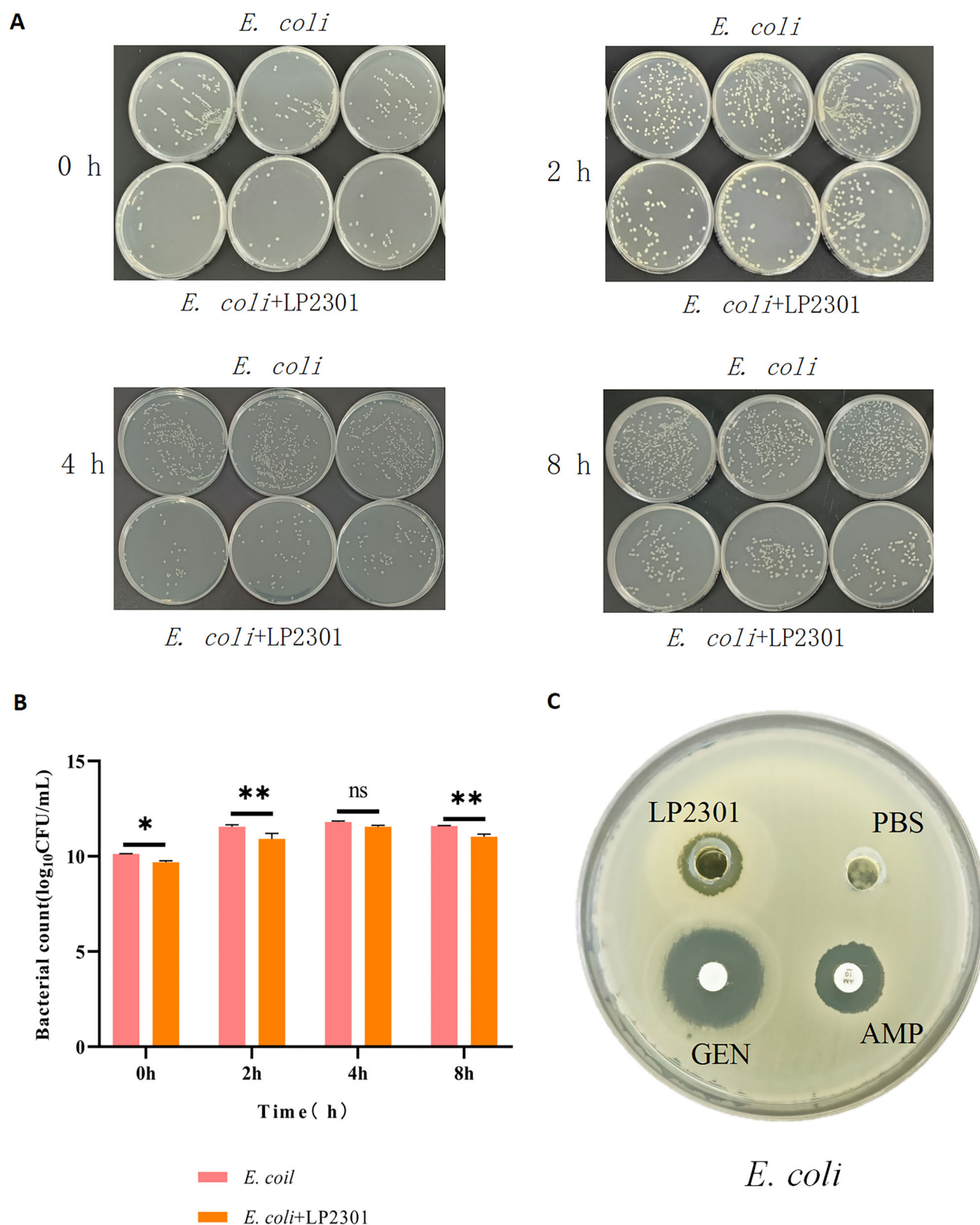


Fig. 4 | The antibacterial activity of LP2301. A Growth condition of *E. coli* co-cultured with LP2301 supernatant; **B** Statistical results of co-culture of *E. coli* and LP2301 supernatant; **C** Antimicrobial halo test of LP2301 supernatant against *E. coli*.

became turbid. Bacterial DNA was extracted, and universal bacterial 16S rRNA gene primers 27F (5'-AGAGTTTGATCCTGGCTCAG-3') and 1492R (5'-GGTACCTTGTTACGACTT-3') were used for PCR amplification. The samples were sent to Sangon Biotech Co., Ltd.

(Shanghai, China) for identification. All *L. plantarum* cultures were incubated statically at 37 °C in an anaerobic jar with an anaerobic culture bag (Mitsubishi Gas Chemical Company) to maintain anaerobic conditions.

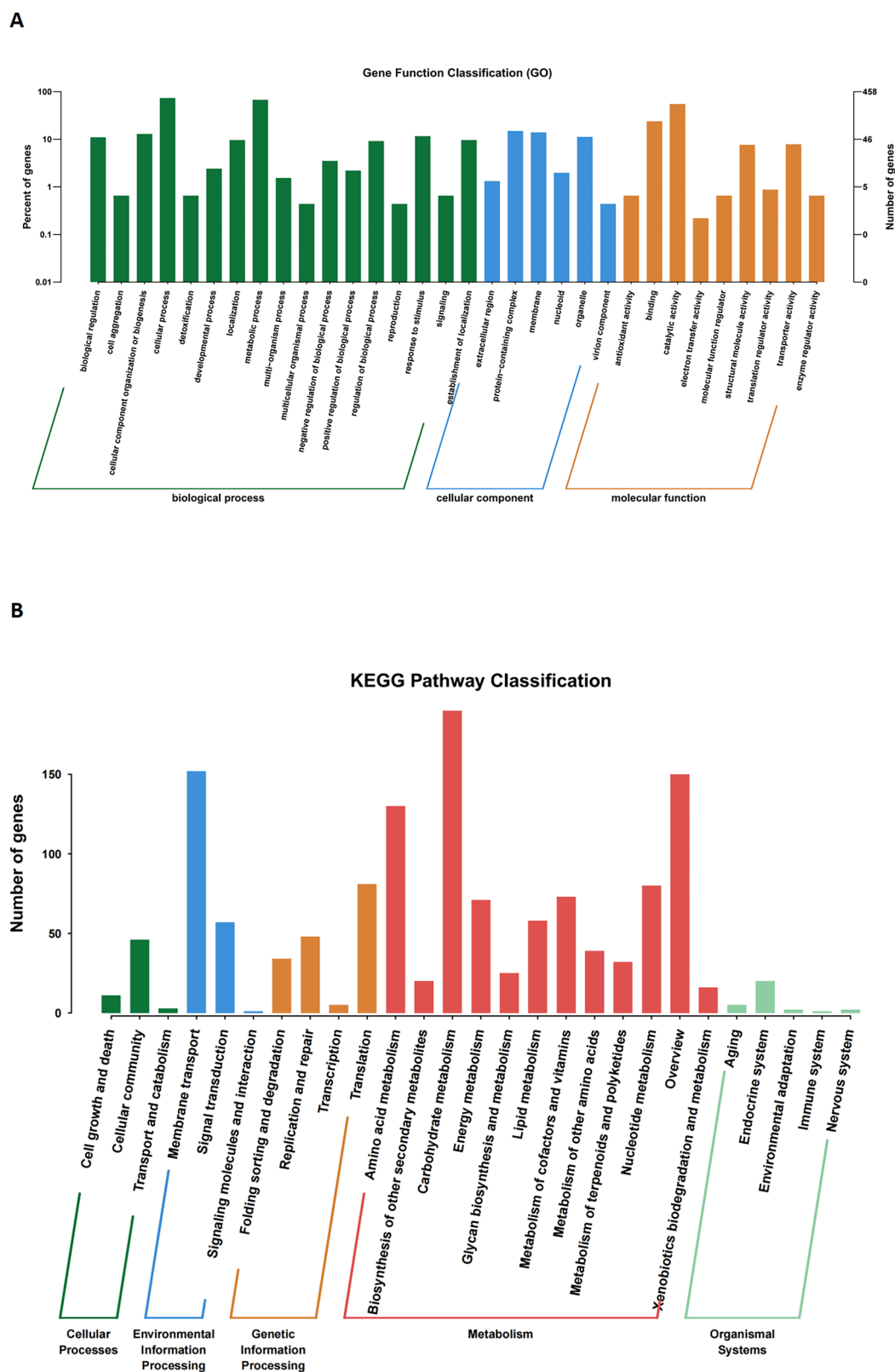


Fig. 5 | Gene enrichment analysis. GO (A) and KEGG (B) gene enrichment analysis.

Biosecurity statement

All clinical isolates were treated strictly in accordance with the Regulations on Biological Safety of Pathogen Microbiology Laboratory by the State Council of the People's Republic of China (000014349/2004-00195). All necessary safety operations were implemented to avoid pathogen transmission and infection.

Gram staining

LP2301 was cultured on MRS nutrient agar (AOBOX, Beijing, China) and incubated upside down overnight at 37 °C. According to the method previously introduced⁵⁷, LP2301 was stained according to the instructions provided with the Gram Staining Kit (Solarbio, Beijing, China).

Table 2 | The annotation results of PHI

PHI	Gene ID	PHI gene	Pathogen species
A0A059ZR97	ctg00006_01536	AKJ44529	<i>Acinetobacter baumannii</i>
A0A069B091	ctg00013_02236	KGD52748	<i>Burkholderia pseudomallei</i>
A0A076ZHT9	ctg00003_01038	AIK80567	<i>Klebsiella pneumoniae</i>
A0A076ZK37	ctg00006_01625; ctg00009_01914; ctg00002_00612	AIK80586	<i>Klebsiella pneumoniae</i>
A0A086IQC3	ctg00004_01210; ctg00002_00590; ctg00001_00372; ctg00001_00254; ctg00010_02008; ctg00022_02761; ctg00007_01646	AIS99989	<i>Klebsiella pneumoniae</i>
A0A097IB07	ctg00015_02429	AIT59279	<i>Riemerella anatipestifer</i>
A0A0B7LYL4	ctg00002_00780	CEV82963	<i>Streptococcus pneumoniae</i>
A0A0B7MDN1	ctg00002_00778	CEV82959	<i>Streptococcus pneumoniae</i>
A0A0D5YKF1	ctg00009_01901; ctg00003_01005	AKA32424	<i>Acinetobacter baumannii</i>
A0A0E0UVG9	ctg00006_01520	AEH92113	<i>Listeria monocytogenes</i>
A0A0F6B0W4	ctg00001_00161	ACY88142	<i>Salmonella enterica</i>
A0A0F6UGK4	ctg00001_00155	CRQ13186	<i>Pseudomonas aeruginosa</i>
A0A0H2UR33	ctg00002_00659	AAK75760	<i>Streptococcus pneumoniae</i>
A0A0H2URB2	ctg00001_00043	AAK75761	<i>Streptococcus pneumoniae</i>
A0A0H2WXT5	ctg00042_03061; ctg00026_02868; ctg00030_02936; ctg00014_02355	AAW37467	<i>Staphylococcus aureus</i>
A0A0H2XG07	ctg00027_02895	ABD21364	<i>Staphylococcus aureus</i>
A0A0H2XGW3	ctg00016_02505	ABD21090	<i>Staphylococcus aureus</i>
A0A0H2XHD4	ctg00025_02835; ctg00001_00354; ctg00013_02252; ctg00004_01172	ABD22043	<i>Staphylococcus aureus</i>
A0A0H2ZQ21	ctg00012_02195; ctg00011_02134	ABJ54846	<i>Streptococcus pneumoniae</i>
A0A0H3HVK0	ctg00016_02470	AFI69833	<i>Burkholderia pseudomallei</i>

Biochemical identification of LP2301

The biochemical characterization of *L. plantarum* LP2301 was conducted according to the instructions of Micro-biochemical identification tubes for bacteria (HopeBio, Qingdao, Shandong, China). The specific indicators measured include oxidase, catalase, fructose, galactose, xylose, D-arabinose, D-melibiose, bile broth, esculin, cellobiose, maltose, mannitol, salicin, sorbitol, sucrose, raffinose, inulin, and lactose.

LP2301 De novo sequencing

Genomic DNA was extracted using the TIANamp Bacteria DNA Kit (Tiangen, Beijing). The whole genome DNA was randomly fragmented to an average size of 200–400 bp. The selected fragments were through end-repair, 3' adenylated, adapter-ligation, and PCR amplification. After purification with the magnetic beads, the library was qualified by the Qubit 4.0 fluorometer, and the length of the library was assessed by the 2% agarose gel electrophoresis. The qualified libraries were sequenced on the MGI DNBSEQ-T7 platform at Sangon Biotech (Shanghai, China). After sequencing, raw reads were filtered via Fastp (v 0.23.0) by removing adapters and low-quality reads, then clean reads were obtained. A genome survey was performed using Jellyfish (v 2.3.0) for k-mer counting and GenomeScope2 for genome size and heterozygosity estimation based on clean reads. Genome assembly was done using SPAdes (v 3.5.0), and the Gapfiller (v 1.11) was used for filling gaps. Functional gene annotation was performed through homology-based searches using DIAMOND (v2.0.8) against multiple specialized databases, including NR (general functions), KEGG (metabolic pathways), VFDB (virulence factors), CARD (antibiotic resistance), CAZy (carbohydrate-active enzymes), PHI-base (host-pathogen interactions), ARGs (antimicrobial resistance genes), TCDB (transporters), and BacMet (biocide and metal resistance). Additionally, Gene Ontology (GO) terms and COG categories were assigned via orthology mapping with eggNOG-mapper (v2.1.4). Protein feature prediction was further conducted to identify transmembrane helices (TMHMM v2.0), signal peptides (SignalP v6.0), lipoproteins (LipoP v1.0), and subcellular localization (ProtCamp v9.0) for comprehensive functional characterization.

Growth curve of LP2301

To better understand the growth characteristics of LP2301, its growth curve was detected by a turbidimeter. A total of 1 mL of MRS broth was added to a sterile turbidimetric tube to zero the turbidity meter, and then the bacterial culture was dripped into the tube until its turbidity reached 0.5 MacFarland Criteria (MCF). After that, 198 μ L of MRS broth was supplied to each well of a 96-well plate, and then added 2 μ L of bacterial culture with a turbidity of 0.5 MCF was added to each well. The 96-well plate was placed in a micro-plate reader at 37 °C, and the OD₆₀₀ was measured once every hour for 48 consecutive hours. This experiment was performed with at least three independent technical replicates. Data are presented as mean $\pm$ SEM.

Acid resistance rest of LP2301

The LP2301 bacterial culture was centrifuged at 8000 $\times$ g for 5 min. Subsequently, it was resuspended using 2 mL of sterile physiological saline. Serial dilution (10-fold) was then carried out, and 10 μ L of the appropriate dilution was transferred to a cell-counting plate to enumerate the number of viable bacteria. Based on this enumeration, the concentration of the original bacterial culture was calculated. According to the determined bacterial culture concentration, an appropriate volume of the bacterial culture was added to 0.1 mol/L HCl-MRS broth medium (pH = 2.5) to regulate the final concentration of the LP2301 bacterial culture to 10⁷ units per milliliter (CFU/mL). After anaerobic incubation at 37 °C for 2 h, samples of the LP2301 bacterial culture were rapidly withdrawn at 0, 0.5, 1, and 2 h. These samples were centrifuged at 8000 $\times$ g for 5 min, the supernatant was discarded, and the pellet was resuspended with an appropriate volume of sterile physiological saline. Serial dilution was performed again, and specific serial dilution concentrations were selected for colony counting. Based on the observation of the number of colonies on plates at various dilutions in the preliminary experiment, dilutions of 10⁻³ and 10⁻⁴ were employed in this experiment. Ten micro-liters of the bacterial suspension at each dilution was evenly spread on MRS plates. A control group was established. After anaerobic incubation at 37 °C for 48–72 h, the number of colonies on the plates at each time point was counted. The concentration of the original bacterial culture was calculated using the formula: concentration of original

Table 3 | Summary of virulence factor annotations

VFDB A	GeneID	VF symbol	VF Information	VF name	VF term
VFG000077	ctg000017_02518	clpP	ATP-dependent Clp protease proteolytic subunit	ClpP	VF0074
VFG000079	ctg000016_02470; ctg000004_01251	clpC	endopeptidase Clp ATP-binding chain C	ClpC	VF0072
VFG000080	ctg000033_02970; ctg000019_02623	clpE	ATP-dependent protease	ClpE	VF0073
VFG000130	ctg000007_01693	algI	alginate o-acetyltransferase AlgI	Alginate	VF0091
VFG000182	ctg000018_02576	xcpR	general secretion pathway protein E	xcp secretion system	VF0084
VFG000320	ctg000015_02404	kdtB	lipopolysaccharide core biosynthesis protein	LPS	VF0056
VFG000344	ctg000001_00042; ctg000019_02627; ctg000011_02122; ctg000006_01527; ctg000019_02626; ctg000002_00781; ctg000003_00856	hitC	iron(III) ABC transporter, ATP-binding protein	HitABC	VF0268
VFG000430	ctg000015_02429	tvIC	Vi polysaccharide biosynthesis protein, epimerase TvIC	Vi antigen	VF0101
VFG000574	ctg000006_01625	mgfB	Mg2+ transport protein	MgtBC	VF0106
VFG000670	ctg000007_01649; ctg000001_00122	gtrB	bactoprenol glucosyl transferase	LPS	VF0124
VFG000907	ctg000013_02266; ctg000001_00372	hlyB	Hemolysin B	Hemolysin	VF0225
VFG000964	ctg000006_01520	hasC	UDP-glucose pyrophosphorylase	Hyaluronic acid capsule	VF0244
VFG001206	ctg000001_00378; ctg000010_01972; ctg000002_00590; ctg000015_02424; ctg000011_02078; ctg000017_02535; ctg000001_00254; ctg000012_02160; ctg000010_02008; ctg000022_02761; ctg000007_01646; ctg000001_00052	fbpC	iron(III) ABC transporter, ATP-binding protein	FbpABC	VF0272
VFG001285	ctg000001_00181	icaA	N-acetylglucosaminyltransferase, involved in polysaccharide intercellular adhesin(PIA) synthesis	Intercellular adhesion proteins	VF0014
VFG001298	ctg000015_02427	cap8B	capsular polysaccharide synthesis enzyme Cap8B	Capsule	VF0003
VFG001373	ctg000004_01070	cps4I	UDP-N-acetylglucosamine-2-epimerase	Capsule	VF0144
VFG001386	ctg000006_01533; ctg000001_00197; ctg000002_00748	phoP	Possible two-component system response transcriptional positive regulator PhoP	PhoP	VF0286
VFG001826	ctg000007_01706	relA	Probable GTP pyrophosphokinase RelA (ATP:GTP 3'-pyrophosphotransferase) (PPGPP synthetase I) (P)PPGPP synthetase) (GTP diphosphokinase)	RelA	VF0287
VFG001855	ctg000006_01545	htpB	Hsp60, 60 K heat shock protein HtpB	Hsp60	VF0159
VFG001859	ctg000001_00155	feoB	ferrous iron transporter B	FeoAB	VF0160

Table 4 | Overview of signal peptide proteins

Organism Type	Gram +
Total Proteins Number	2981
Signal Proteins from SignalP-TM	150
Signal Proteins from SignalP-noTM	No Such Protein from Gram+ bacteria
Total Sigan Proteins	150
Signal Protein Ratio (%)	5.03

bacterial culture (CFU/mL) = actual number of colonies (CFU)/sampling volume (mL) × dilution factor. All experiments were performed with at least three independent technical replicates. Data are presented as mean ± SEM.

Bile salt tolerance test of LP2301

The LP2301 bacterial culture was centrifuged at 8000 × g for 5 min. Subsequently, it was resuspended with 2 mL of sterile physiological saline, subjected to serial dilution (10-fold). Then, 10 µL of the appropriate dilution was transferred to a cell-counting plate to enumerate the number of viable bacteria. Based on the obtained results, the concentration of the original bacterial culture was calculated. According to the determined bacterial culture concentration, 10 µL of the bacterial culture was added to MRS broth medium with a porcine bile salt concentration of 0.3%, ensuring that the final concentration of the LP2301 bacterial culture was adjusted to 10⁷ CFU/mL. After anaerobic incubation at 37 °C for 4 h, the LP2301 bacterial culture was rapidly sampled at 0, 2, and 4 h. The sampled solutions were centrifuged at 8000 × g for 5 min, the supernatants were discarded, and the pellets were resuspended with an appropriate volume of sterile physiological saline. Serial dilution was then carried out, and the serially diluted solutions were used for colony counting. Based on the observation of the colony numbers on plates at various dilutions in the preliminary experiment, dilutions of 10⁻³ and 10⁻⁴ were selected for this experiment. For each dilution, 10 µL of the bacterial suspension was evenly spread on MRS plates. A control group was also established. After anaerobic incubation at 37 °C for 48–72 h, the colony numbers on the plates at each time point were counted. The concentration of the original bacterial culture was calculated according to the colony numbers using the formula: Original bacterial culture concentration (CFU/mL) = actual number of colonies (CFU)/sampling volume (mL) × dilution factor. All experiments were performed with at least three independent technical replicates. Data are presented as mean ± SEM.

Antimicrobial activity assay of LP2301

The sensitivity of *E. coli* to LP2301 supernatant was assessed as previously described⁵⁸, which is a co-incubation assay measuring reduction in pathogen CFU at fixed time points (0, 2, 4, and 8 h). LP2301 was cultured in MRS broth at 37 °C for 24 h under anaerobic conditions. The culture was centrifuged at 10,000 × g for 10 min at 4 °C. The supernatant was filtered through a 0.22 µm sterile filter (Millipore) to remove bacterial cells. For protease treatment, the supernatant was incubated with 1 mg/mL proteinase K (Sigma) at 37 °C for 2 h, followed by heat inactivation of the enzyme. Subsequently, 50 µL of the supernatant was extracted and uniformly spread onto three sterile LB agar plates. The plates were placed on the clean bench for 15 minutes to enable the bacterial culture to permeate into the culture medium. The *E. coli* solution was subjected to gradient dilution, and 50 µL of the diluted bacterial culture was extracted for inoculation onto plates pre-inoculated with LP2301 supernatant. Simultaneously, 50 µL of the *E. coli* suspension was taken for inoculation onto fresh LB plates, which would function as the control group. After standing for 15 minutes, all the plates were inverted and incubated at a 37 °C incubator for 24 h before counting. Gentamicin (GM) and ampicillin (AMP) served as positive controls. All experiments were performed with at least three independent technical replicates. Data are presented as mean ± SEM. The antibacterial zone test of the supernatant of LP2301 against *E. coli*, *S. aureus*, *S. typhimurium*, and *B.*

cereus was conducted. The supernatant of LP2301 after Protease K treatment was also tested against *E. coli* and *S. aureus*.

Statistical analysis

Each experiment was replicated at least three times, yielding comparable results. All data were analyzed by GraphPad Prism version 10. The data met the assumptions of normal distribution and homogeneity of variance (verified by the Shapiro-Wilk test and Levene test, $P > 0.05$), thus two-way ANOVA was selected, with treatment and time as fixed effects. If the overall analysis showed significant differences ($P < 0.05$), the Sidak test was used for post-hoc multiple comparisons to determine the significant differences between different treatment groups at specific time points. The significance levels were set as follows: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Data availability

No datasets were generated or analyzed during the current study.

Received: 4 February 2026; Accepted: 15 May 2026;

Published online: 27 May 2026

References

- Shani, L. & Nissan, I. Editorial: Antibiotics overuse as the driving force behind antimicrobial resistance. *Front. Cell Infect. Microbiol.* **14**, <https://doi.org/10.3389/fcimb.2024.1373846>.
- Bauer, K. A., Kullar, R., Gilchrist, M. & File, T. Antibiotics and adverse events: the role of antimicrobial stewardship programs in 'doing no harm. *Curr. Opin. Infect. Dis.* **32**, 553–558, <https://doi.org/10.1097/Qco.0000000000000604> (2019).
- Shallcross, L. J. & Davies, D. S. C. Antibiotic overuse: a key driver of antimicrobial resistance. *Brit. J. Gen. Pr.* **64**, 604–605, <https://doi.org/10.3399/bjgp14X682561> (2014).
- Enshaie, E., Nigam, S., Patel, S. & Rai, V. Livestock antibiotics use and antimicrobial resistance. *Antibiotics* **14**, <https://doi.org/10.3390/antibiotics14060621>.
- He, L. X. et al. Antibiotics, antibiotic resistance genes and microbial community in grouper mariculture. *Sci. Total Environ.* **808**, <https://doi.org/10.1016/j.scitotenv.2021.152042>.
- Alvarado, A. C., Chekabab, S. M., Predicala, B. Z. & Korber, D. R. Impact of raised without antibiotics measures on antimicrobial resistance and prevalence of pathogens in sow barns. *Antibiotics* **11**, <https://doi.org/10.3390/antibiotics11091221>.
- Ramirez, J. et al. Antibiotics as major disruptors of gut microbiota. *Front. Cell Infect. Microbiol.* **10**, <https://doi.org/10.3389/fcimb.2020.572912>.
- Kim, S., Covington, A. & Pamer, E. G. The intestinal microbiota: Antibiotics, colonization resistance, and enteric pathogens. *Immunol. Rev.* **279**, 90–105, <https://doi.org/10.1111/imr.12563> (2017).
- Li, W. Z. et al. Vertical transmission of gut microbiome and antimicrobial resistance genes in infants exposed to antibiotics at birth. *J. Infect. Dis.* **224**, 1236–1246, <https://doi.org/10.1093/infdis/jiaa155> (2021).
- Nogacka, A. et al. Impact of intrapartum antimicrobial prophylaxis upon the intestinal microbiota and the prevalence of antibiotic resistance genes in vaginally delivered full-term neonates. *Microbiome* **5**, <https://doi.org/10.1186/s40168-017-0313-3>.
- Gao, P. F. et al. Feed-additive probiotics accelerate yet antibiotics delay intestinal microbiota maturation in broiler chicken. *Microbiome* **5**, <https://doi.org/10.1186/s40168-017-0315-1>.
- Estany-Gestal, A., Salgado-Barreira, A. & Vazquez-Lago, J. M. Antibiotic use and antimicrobial resistance: a global public health crisis. *Antibiotics* **13**, <https://doi.org/10.3390/antibiotics13090900>.
- Brüssow, H. The antibiotic resistance crisis and the development of new antibiotics. *Microb. Biotechnol.* **17**, <https://doi.org/10.1111/1751-7915.14510>.

14. Aghamohammad, S. & Rohani, M. Antibiotic resistance and the alternatives to conventional antibiotics: The role of probiotics and microbiota in combating antimicrobial resistance. *Microbiol. Res.* **267**, <https://doi.org/10.1016/j.micres.2022.127275>.
15. Meng, F. Q. et al. Plantaricin A. Derived from *Lactiplantibacillus plantarum*, reduces the intrinsic resistance of gram-negative bacteria to hydrophobic antibiotics. *Appl. Environ. Microbiol.* **88**, <https://doi.org/10.1128/aem.00371-22>.
16. Ouwehand, A. C., Forssten, S., Hibberd, A. A., Lyra, A. & Stahl, B. Probiotic approach to prevent antibiotic resistance. *Ann. Med.* **48**, 246–255, <https://doi.org/10.3109/07853890.2016.1161232> (2016).
17. Sabo, S. S., Converti, A., Ichiwaki, S. & Oliveira, R. P. S. Bacteriocin production by *Lactobacillus plantarum* ST16Pa in supplemented whey powder formulations. *J. Dairy Sci.* **102**, 87–99, <https://doi.org/10.3168/jds.2018-14881> (2019).
18. Guimaraes, A., Santiago, A., Teixeira, J. A., Venâncio, A. & Abrunhosa, L. Anti-aflatoxigenic effect of organic acids produced by *Lactobacillus plantarum*. *Int. J. Food Microbiol.* **264**, 31–38, <https://doi.org/10.1016/j.ijfoodmicro.2017.10.025> (2018).
19. Sabo, S. D., Vitolo, M., González, J. M. D. & Oliveira, R. P. D. Overview of *Lactobacillus plantarum* as a promising bacteriocin producer among lactic acid bacteria. *Food Res. Int.* **64**, 527–536, <https://doi.org/10.1016/j.foodres.2014.07.041> (2014).
20. Mirsalami, S. M. & Mirsalami, M. Leveraging *Lactobacillus plantarum* probiotics to mitigate diarrhea and Salmonella infections in broiler chickens. *AMB Express* **14**, <https://doi.org/10.1186/s13568-024-01792-3>.
21. Arena, M. P. et al. Use of *Lactobacillus plantarum* Strains as a Bio-Control Strategy against Food-Borne Pathogenic Microorganisms. *Front. Microbiol.* **7**, <https://doi.org/10.3389/fmicb.2016.00464>.
22. Wang, W. W. et al. Screening of *Lactobacillus plantarum* subsp. *plantarum* with potential probiotic activities for inhibiting ETEC K88 in Weaned Piglets. *Molecules* **25**, <https://doi.org/10.3390/molecules25194481>.
23. Thakur, B. et al. Unveiling the antimicrobial and antibiofilm potential of biosurfactant produced by newly isolated *Lactiplantibacillus plantarum* strain 1625. *Front. Microbiol.* **15**, <https://doi.org/10.3389/fmicb.2024.1459388>.
24. Khalfallah, G., Gartzén, R., Möller, M., Heine, E. & Lütticken, R. A new approach to harness probiotics against common bacterial skin pathogens: towards living antimicrobials. *Probiot. Antimicrob. Proteins* **13**, 1557–1571, <https://doi.org/10.1007/s12602-021-09783-7> (2021).
25. Lee, C. S. & Kim, S. H. Anti-inflammatory and Anti-osteoporotic Potential of *Lactobacillus plantarum* A41 and *L. fermentum* SRK414 as Probiotics. *Probiot. Antimicrob. Proteins* **12**, 623–634, <https://doi.org/10.1007/s12602-019-09577-y> (2020).
26. Sun, Y. F. et al. Study on the alleviative effect of *Lactobacillus plantarum* on *Eimeria falciformis* infection. *Infect. Immun.* **92**, <https://doi.org/10.1128/iai.00130-24>.
27. Wang, C., Chuprom, J., Wang, Y. & Fu, L. Beneficial bacteria for aquaculture: nutrition, bacteriostasis and immunoregulation. *J. Appl. Microbiol.* **128**, 28–40, <https://doi.org/10.1111/jam.14383> (2020).
28. Kuda, T., Yokota, Y., Haraguchi, Y., Takahashi, H. & Kimura, B. Susceptibility of gut indigenous lactic acid bacteria in BALB/c mice to oral administered (vol 70, pg 53, 2018). *Int. J. Food Sci. Nutr.* **70**, <https://doi.org/10.1080/09637486.2018.1492364>.
29. Heeney, D. D. et al. *Lactobacillus plantarum* bacteriocin is associated with intestinal and systemic improvements in diet-induced obese mice and maintains epithelial barrier integrity. *Gut Microbes* **10**, 382–397, <https://doi.org/10.1080/19490976.2018.1534513> (2019).
30. Chen, Y. et al. Ameliorates colorectal cancer by ameliorating the intestinal barrier through the CLA-PPAR- γ axis. *J. Agr. Food Chem.* **72**, 19766–19785, <https://doi.org/10.1021/acs.jafc.4c02824> (2024).
31. Zhou, X. T. et al. Exopolysaccharides from *Lactobacillus plantarum* NCU116 Regulate Intestinal Barrier Function via STAT3 Signaling Pathway. *J. Agr. Food Chem.* **66**, 9719–9727, <https://doi.org/10.1021/acs.jafc.8b03340> (2018).
32. Xiong, D. et al. Screening of *Lactobacillus plantarum* LPH10 and its anti-inflammatory properties: acid and bile salt tolerance, bacteriostatic activity and regulation of NF- κ B/IRF pathways. *Probiot. Antimicrob. Proteins* **18**, 2732–2745, <https://doi.org/10.1007/s12602-025-10725-w> (2026).
33. Bachtarzi, N. et al. Selection of Exopolysaccharide-Producing *Lactobacillus plantarum* (*Lactiplantibacillus plantarum*) Isolated from Algerian Fermented Foods for the Manufacture of Skim-Milk Fermented Products. *Microorganisms* **8**, <https://doi.org/10.3390/microorganisms8081101>.
34. Liang, J. Q. et al. Enhancing fermented vegetable flavor with *Lactobacillus plantarum* and *Rhodotorula mucilaginosa*. *Food Res. Int.* **200**, <https://doi.org/10.1016/j.foodres.2024.115500>.
35. Ventimiglia, G. et al. Codominance of *Lactobacillus plantarum* and obligate heterofermentative lactic acid bacteria during sourdough fermentation. *Food Microbiol.* **51**, 57–68, <https://doi.org/10.1016/j.fm.2015.04.011> (2015).
36. Oh, Y. J. et al. *Lactobacillus plantarum* PMO 08 as a Probiotic Starter Culture for Plant-Based Fermented Beverages. *Molecules* **25**, <https://doi.org/10.3390/molecules25215056>.
37. Xiong, D. et al. Screening of *Lactobacillus plantarum* LPH10 and its anti-inflammatory properties: acid and bile salt tolerance, bacteriostatic activity and regulation of NF- κ B/IRF pathways. *Probiotics Antimicrob. Proteins*. <https://doi.org/10.1007/s12602-025-10725-w>.
38. Elmodallal, H., Siçramaz, H. & Cosansu, S. Terebinth (*Pistacia terebinthus*) seed milk as a growth medium and delivery matrix for *Lactobacillus acidophilus* and *Lactiplantibacillus plantarum*. *J. Sci. Food Agr.* <https://doi.org/10.1002/jsfa.70340>.
39. Heunis, T., Deane, S., Smit, S. & Dicks, L. M. T. Proteomic profiling of the acid stress response in *Lactobacillus plantarum* 423. *J. Proteome Res.* **13**, 4028–4039, <https://doi.org/10.1021/pr500353x> (2014).
40. Ricciardi, A. et al. Genotypic diversity of stress response in *Lactobacillus plantarum*, *Lactobacillus paraplantarum* and *Lactobacillus pentosus*. *Int. J. Food Microbiol.* **157**, 278–285, <https://doi.org/10.1016/j.ijfoodmicro.2012.05.018> (2012).
41. Huang, R. H. et al. Physiological and transcriptional responses and cross protection of ZDY2013 under acid stress. *J. Dairy Sci.* **99**, 1002–1010, <https://doi.org/10.3168/jds.2015-9993> (2016).
42. Hamon, E. et al. Comparative proteomic analysis of *Lactobacillus plantarum* for the identification of key proteins in bile tolerance. *BMC Microbiol.* **11**, <https://doi.org/10.1186/1471-2180-11-63> (2011).
43. Ahire, J. J., Jakkamsetty, C., Kashikar, M. S., Lakshmi, S. G. & Madempudi, R. S. In vitro evaluation of probiotic properties of *Lactobacillus plantarum* UBLP40 isolated from traditional indigenous fermented food. *Probiot. Antimicrob. Proteins* **13**, 1413–1424, <https://doi.org/10.1007/s12602-021-09775-7> (2021).
44. Feng, X. et al. Bile salt hydrolase of *Lactiplantibacillus plantarum* plays important roles in amelioration of DSS-induced colitis. *Is Science* **26**, <https://doi.org/10.1016/j.isci.2023.106196>.
45. Shao, J. L. et al. *bsh 1* Gene of *Lactobacillus plantarum* AR113 Plays an Important Role in Ameliorating Western Diet-Aggravated Colitis. *J. Agr. Food Chem.* **71**, 9337–9348, <https://doi.org/10.1021/acs.jafc.2c08631> (2023).
46. Prete, R. et al. Food-Associated *Lactobacillus plantarum* and yeasts inhibit the genotoxic effect of 4-Nitroquinoline-1-Oxide. *Front. Microbiol.* **8**, <https://doi.org/10.3389/fmicb.2017.02349>.
47. Prete, R. et al. Beneficial bile acid metabolism from *Lactobacillus plantarum* of food origin. *Sci. Rep.-Uk* **10**, <https://doi.org/10.1038/s41598-020-58069-5>.

48. Bai, X. Y. et al. *Lactiplantibacillus plantarum* Lp20 alleviates high fat diet-induced obesity in mice via its bile salt hydrolase activity. *Nutrients* **17**, <https://doi.org/10.3390/nu17223555>.
49. Zhong, H. et al. *Lactobacillus plantarum* ZJUF2 prevents high fat diet-induced insulin resistance in association with modulation of the gut microbiota. *Front. Nutr.* **8**, <https://doi.org/10.3389/fnut.2021.754222>.
50. Mao, Y., Zhang, X. J. & Xu, Z. H. Identification of antibacterial substances of *Lactobacillus plantarum* DY-6 for bacteriostatic action. *Food Sci. Nutr.* **8**, 2854–2863, <https://doi.org/10.1002/fsn3.1585> (2020).
51. Rao, K. P. et al. Antiaflatoxigenic Potential of Cell-Free Supernatant from *Lactobacillus plantarum* MYS44 Against *Aspergillus parasiticus*. *Probiot. Antimicrob. Proteins* **11**, 55–64, <https://doi.org/10.1007/s12602-017-9338-y> (2019).
52. Ryu, E. H., Yang, E. J., Woo, E. R. & Chang, H. C. Purification and characterization of antifungal compounds from *Lactobacillus plantarum* HD1 isolated from kimchi. *Food Microbiol.* **41**, 19–26, <https://doi.org/10.1016/j.fm.2014.01.011> (2014).
53. Kim, S. W. et al. Potential of Cell-Free Supernatant from *Lactobacillus plantarum* NIBR97, including novel Bacteriocins, as a natural alternative to chemical disinfectants. *Pharmaceuticals* **13**, <https://doi.org/10.3390/ph13100266>.
54. De Giani, A. et al. Identification of a bacteriocin-like compound from *Lactobacillus plantarum* with antimicrobial activity and effects on normal and cancerogenic human intestinal cells. *AMB Express* **9**, <https://doi.org/10.1186/s13568-019-0813-6>.
55. Hu, C. H., Ren, L. Q., Zhou, Y. & Ye, B. C. Characterization of antimicrobial activity of three *Lactobacillus plantarum* strains isolated from Chinese traditional dairy food. *Food Sci. Nutr.* **7**, 1997–2005, <https://doi.org/10.1002/fsn3.1025> (2019).
56. Rao, W. et al. Antibacterial mechanism of metabolites of *Lactobacillus plantarum* against *Pseudomonas lundensis* and their application in dry-aged beef preservation. *Food Chem.* **460**, <https://doi.org/10.1016/j.foodchem.2024.140463>.
57. Shan, Q. et al. *Lactobacillus rhamnosus* GR-1 attenuates foodborne induced NLRP3 inflammasome activity in bovine mammary epithelial cells by protecting intercellular tight junctions. *J. Anim. Sci. Biotechnol.* **13**, <https://doi.org/10.1186/s40104-022-00752-w>.
58. Du, M. Z. et al. Antimicrobial Effect of *Zophobas morio* Hemolymph against Bovine Mastitis Pathogens. *Microorganisms* **8**, <https://doi.org/10.3390/microorganisms8101488>.

Acknowledgements

This work was supported by the National Key R&D program of China (Grant No. 2023YFD1801100), the program for Postdoctoral of Heilongjiang

Province (Project no. LBH-Z23085), and the Excellent Youth Project of Heilongjiang Provincial Natural Science Foundation (Project No. YQ2024C020).

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H.F.: Writing-review & editing, Software, Methodology, Formal analysis. Q.L.: Writing-review & editing, Methodology, Investigation, Formal analysis. M.S.: Methodology, Investigation, Formal analysis. Y.S.: Methodology, Investigation. Y.W.: Software, Methodology. N.L.: Writing-original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Supervision, Project administration.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41538-026-00908-2>.

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